Stateful Web Application

Total points 20/20

Assessment covering session tracking, cookies, URL rewriting, hidden fields, HttpSession, servlet chaining, filters, listeners, ServletConfig/ServletContext, annotations, and basics of authentication & authorization. 20 multiple-choice questions and 2 written questions.

✓ Which of the following is NOT a session tracking technique in web applications? 1/1

- ○ Cookies
- ○ URL Rewriting
- ○ Hidden Fields
- ◉ HTML5 Canvas ✓

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✓ What is the primary purpose of a cookie? 1/1

- ○ To store server-side session objects
- ◉ To maintain state on the client between HTTP requests ✓
- ○ To rewrite URLs automatically
- ○ To define servlet mappings

✓ Which of the following attributes of javax.servlet.http.Cookie controls the lifetime of the cookie on the client? 1/1

- ○ domain
- ○ path
- ◉ maxAge ✓
- ○ secure

✓ URL rewriting for session tracking typically appends what to the URL? 1/1

- ○ A cookie header
- ◉ Session identifier as a path or query parameter ✓
- ○ Encrypted payload in the request body
- ○ A custom HTTP method

✓ Hidden fields are most appropriate for session tracking in which scenario? 1/1

- ○ When tracking state across multiple tabs
- ◉ When maintaining state across successive form submissions in a single browser session ✓
- ○ When the client should be completely stateless
- ○ When storing large binary data on the client

✓ Which class provides server-side session management in Java EE (Servlet API)? 1/1

- ◉ javax.servlet.http.HttpSession ✓
- ○ javax.servlet.SessionBean
- ○ java.net.Session
- ○ javax.servlet.http.CookieSession

✓ Which HttpSession method is used to store an attribute in the session? 1/1

- ◉ setAttribute(name, value) ✓
- ○ put(name, value)
- ○ addAttribute(name, value)
- ○ store(name, value)

✓ Which session tracking technique will still work if the client has disabled cookies but JavaScript and links are available? 1/1

- ○ Cookies
- ◉ URL Rewriting ✓
- ○ HTTP Basic Auth
- ○ Local Storage

✓ Which of the following is a major drawback of URL rewriting? 1/1

- ○ It cannot be used with HTML forms
- ◉ Session id appears in URL, which may be exposed in logs or referrals ✓
- ○ It requires cookies to be enabled
- ○ It stores too much data on the client

✓ Servlet chaining is best described as: 1/1

- ◉ Using multiple servlets sequentially to process one request/response ✓
- ○ Running servlets in separate JVMs and merging responses
- ○ Replacing all servlets with JSP pages
- ○ Executing filters after the response is committed

✓ What is the primary role of a servlet Filter? 1/1

- ○ To map URLs to servlets
- ◉ To intercept and possibly modify request and/or response before reaching a servlet ✓
- ○ To create new HttpSession objects
- ○ To compile JSP into servlet classes

✓ Which Filter method is called once when the filter is placed into service? 1/1

- ○ doFilter
- ◉ init ✓
- ○ destroy
- ○ service

✓ ServletContext is typically used for which purpose? 1/1

- ○ Storing per-user data
- ◉ Storing application-wide initialization parameters and resources ✓
- ○ Replacing HttpSession functionality
- ○ Handling low-level socket connections

✓ ServletConfig differs from ServletContext in that ServletConfig: 1/1

- ○ Is shared across the whole web application
- ◉ Provides initialization parameters specific to a single servlet ✓
- ○ Is used to manage user sessions
- ○ Is not available to servlets

✓ Which of the following is an example of a servlet listener? 1/1

- ◉ HttpSessionListener ✓
- ○ ServletFilter
- ○ CookieManager
- ○ SocketListener

✓ What does the @WebServlet annotation do? 1/1

- ○ Defines an EJB component
- ◉ Declares a servlet and its URL mapping without web.xml ✓
- ○ Automatically starts an HTTP server
- ○ Encrypts session data

✓ Which authentication mechanism involves prompting the browser for a username and password and sending them with each request encoded in headers? 1/1

- ○ Form-based authentication
- ○ Digest access authentication
- ◉ HTTP Basic authentication ✓
- ○ Token-based OAuth

✓ In role-based authorization, access is granted primarily based on: 1/1

- ○ The client IP address
- ◉ The user's assigned roles or privileges ✓
- ○ The presence of cookies
- ○ The request method (GET/POST)

✓ Which of these is a recommended practice for securing session cookies? 1/1

- ◉ Setting HttpOnly and Secure flags ✓
- ○ Using predictable session ids
- ○ Sending session id in every URL
- ○ Storing the session id in localStorage

✓ Which listener would you implement to get notifications when a session is created or destroyed? 1/1

- ○ ServletContextListener
- ○ ServletRequestListener
- ◉ HttpSessionListener ✓
- ○ FilterRegistrationListener

Explain the main differences (advantages and disadvantages) between cookies, URL rewriting, and HttpSession for session tracking. Provide at least two points for each technique.

Design a simple secure login flow for a web application using servlets: describe the steps from login form submission to session creation, including how you would protect the session and what checks you would perform to prevent common attacks (e.g., session fixation, XSS).

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